CSA 4302 - INTERNET PROGRAMMING

LAB EXPERIMENT 03:

AIM : To design and develop an Online Bus Ticket Booking System that enables users to register, log in, search for buses, view bus categories and available seats, book tickets, and cancel reservations. The system provides an administrator module to manage bus information, bus categories, registered users, comments, news, and booking details. It also includes password management, report generation, and user authentication to ensure efficient and convenient bus reservation and administration.

CODE:

```
<!doctype html>
<html lang="en">
<head>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>Cornerstone Library</title>
  <style>
    :root {
      color-scheme: light;
      --ink: #24312d;
      --muted: #68756f;
      --paper: #f7f8f4;
      --card: #fff;
      --green: #27634f;
      --green-dark: #194b3a;
      --line: #e3e9e3;
      --accent: #eaf2ed;
      --danger: #a43c34;
      font-family: Inter, ui-sans-serif, system-ui, -apple-system, BlinkMacSystemFont,
"Segoe UI", sans-serif;
    }
    * { box-sizing: border-box; }
    body { margin: 0; background: var(--paper); color: var(--ink); }
    button, input, select, textarea { font: inherit; }
    button { cursor: pointer; }
    header { background: var(--card); border-bottom: 1px solid var(--line); }
    .header-inner { max-width: 1100px; margin: auto; padding: 16px 24px; display:
flex; align-items: center; justify-content: space-between; gap: 18px; }
    .brand { font-size: 1.15rem; font-weight: 750; color: var(--green-dark); letter-
spacing: -.03em; }
    .brand span { color: #9a7952; }
    nav { display: flex; gap: 7px; flex-wrap: wrap; }
```

```
    nav button, .link-button { background: transparent; border: 0; color:
var(--muted); padding: 9px 11px; border-radius: 8px; }
    nav button:hover, nav button.active { background: var(--accent); color: var(--green-dark); }
    .header-actions { display: flex; align-items: center; gap: 10px; }
    .user-label { color: var(--muted); font-size: .9rem; }
    .container { max-width: 1100px; padding: 36px 24px 60px; margin: auto; }
    .hero { display: grid; grid-template-columns: 1.4fr .8fr; gap: 28px; align-items:
center; background: #eaf1e9; border-radius: 18px; padding: 34px; margin-bottom: 30px;
}
    .eyebrow { color: var(--green); text-transform: uppercase; letter-spacing: .12em;
font-size: .75rem; font-weight: 700; }
    h1 { font-size: clamp(2rem, 5vw, 3.2rem); line-height: 1.05; letter-spacing: -
.05em; margin: 10px 0 14px; }
    h2 { letter-spacing: -.03em; margin: 0 0 8px; }
    h3 { margin: 0 0 8px; }
    p { color: var(--muted); line-height: 1.6; }
    .hero p { max-width: 530px; margin: 0 0 20px; }
    .hero-art { min-height: 185px; border-radius: 14px; background:
linear-gradient(145deg, #cbdccd, #e8d9bd); display: grid; place-items: center; font-
size: 5rem; }
    .button { border: 1px solid var(--green); border-radius: 8px; padding: 9px 13px;
background: var(--green); color: white; font-weight: 650; }
    .button:hover { background: var(--green-dark); }
    .button.secondary { color: var(--green-dark); background: white; }
    .button.secondary:hover { background: var(--accent); }
    .button.danger { background: white; color: var(--danger); border-color: #e5b9b5; }
    .button.small { padding: 6px 9px; font-size: .86rem; }
    .section-heading { display: flex; justify-content: space-between; align-items:
end; gap: 14px; margin: 28px 0 15px; }
    .section-heading p { margin: 4px 0 0; }
    .grid { display: grid; grid-template-columns: repeat(3, 1fr); gap: 16px; }
    .card { background: var(--card); border: 1px solid var(--line); border-radius:
12px; padding: 18px; }
    .book-card { display: flex; flex-direction: column; min-height: 195px; }
    .book-card .meta { color: var(--muted); font-size: .9rem; }
    .book-card .availability { margin-top: auto; padding-top: 14px; display: flex;
justify-content: space-between; align-items: center; gap: 12px; }
    .badge { border-radius: 99px; background: var(--accent); color: var(--green-dark);
padding: 4px 9px; font-size: .78rem; font-weight: 650; }
    .badge.unavailable { color: var(--danger); background: #f9e9e7; }
    .toolbar { display: flex; gap: 10px; margin-bottom: 16px; }
    input, select, textarea { width: 100%; padding: 10px 11px; border-radius: 8px;
border: 1px solid #d7dfd8; background: white; color: var(--ink); }
    input:focus, select:focus, textarea:focus { outline: 2px solid #bad5c6; border-
color: var(--green); }
    label { display: grid; gap: 6px; color: #43514b; font-size: .9rem; font-weight:
600; }
    form { display: grid; gap: 12px; }
```

```
    .form-grid { display: grid; grid-template-columns: repeat(2, 1fr); gap: 12px; }
    .stack { display: grid; gap: 16px; }
    .table-wrap { overflow-x: auto; }
    table { border-collapse: collapse; width: 100%; text-align: left; font-size:
.91rem; }
    th, td { padding: 11px 10px; border-bottom: 1px solid var(--line); vertical-align:
middle; }
    th { color: var(--muted); font-weight: 650; font-size: .82rem; }
    .empty { padding: 26px; text-align: center; color: var(--muted); background:
white; border: 1px dashed #cdd8cf; border-radius: 10px; }
    .notice { padding: 12px 14px; border-radius: 8px; background: var(--accent);
color: var(--green-dark); margin-bottom: 14px; }
    .notice.error { background: #f9e9e7; color: var(--danger); }
    .fine { font-weight: 700; color: var(--danger); }
    .hidden { display: none !important; }
    .modal { position: fixed; inset: 0; z-index: 3; background: #15231dcc; display:
grid; place-items: center; padding: 18px; }
    .modal-card { width: min(440px, 100%); background: white; border-radius: 14px;
padding: 24px; box-shadow: 0 16px 60px #0003; }
    .modal-head { display: flex; justify-content: space-between; align-items: center;
margin-bottom: 16px; }
    .close { border: 0; background: transparent; font-size: 1.4rem; color:
var(--muted); }
    footer { border-top: 1px solid var(--line); padding: 23px; text-align: center;
color: var(--muted); font-size: .88rem; }
    @media (max-width: 760px) {
      .header-inner { align-items: flex-start; flex-wrap: wrap; }
      .header-actions { margin-left: auto; }
      nav { order: 3; width: 100%; }
      .container { padding: 25px 16px 45px; }
      .hero { grid-template-columns: 1fr; padding: 25px; }
      .hero-art { min-height: 130px; }
      .grid { grid-template-columns: repeat(2, 1fr); }
    }
    @media (max-width: 520px) {
      .grid, .form-grid { grid-template-columns: 1fr; }
      .toolbar { flex-direction: column; }
      .header-inner { padding: 13px 16px; }
      nav { overflow-x: auto; flex-wrap: nowrap; }
      nav button { white-space: nowrap; }
      .user-label { display: none; }
    }
  </style>
</head>
<body>
  <header>
    <div class="header-inner">
      <div class="brand">Cornerstone <span>Library</span></div>
```

```
        <nav aria-label="Main navigation">
          <button class="active" data-page="home">Home</button>
          <button data-page="catalog">Book catalog</button>
          <button data-page="materials">Study materials</button>
          <button data-page="dashboard" class="hidden" id="dashboard-nav">My
account</button>
          <button data-page="admin" class="hidden" id="admin-nav">Librarian</button>
          <button data-page="contact">Contact</button>
        </nav>
        <div class="header-actions">
          <span class="user-label" id="user-label"></span>
          <button class="button secondary small" id="login-open">Log in</button>
          <button class="button small hidden" id="logout">Log out</button>
        </div>
      </div>
    </header>
    <main class="container">
      <div id="message" class="notice hidden" role="status"></div>
      <section id="page-home" class="page">
        <div class="hero">
          <div>
            <div class="eyebrow">Your campus library</div>
            <h1>Find your next<br>great read.</h1>
            <p>Search the library catalog, explore study materials, and keep track of
the books you borrow—all in one place.</p>
            <button class="button" data-go="catalog">Explore the catalog</button>
          </div>
          <div class="hero-art" aria-hidden="true">📚</div>
        </div>
        <div class="section-heading">
          <div><h2>Popular books</h2><p>A few picks from our library shelves.</p></div>
          <button class="link-button" data-go="catalog">View all books →</button>
        </div>
        <div id="featured-books" class="grid"></div>
      </section>
      <section id="page-catalog" class="page hidden">
        <div class="section-heading"><div><div class="eyebrow">Browse</div><h2>Book
catalog</h2><p>Search by title, author, or subject.</p></div></div>
        <div class="toolbar"><input id="book-search" type="search" placeholder="Search
books…" aria-label="Search books"></div>
        <div id="book-results" class="grid"></div>
      </section>
      <section id="page-materials" class="page hidden">
        <div class="section-heading"><div><div class="eyebrow">For your
studies</div><h2>Study materials</h2><p>Find resources by academic year and
semester.</p></div></div>
        <div class="toolbar">
```

```
        <select id="material-year" aria-label="Filter by year"><option value="">All years</option></select>
        <select id="material-semester" aria-label="Filter by semester"><option value="">All semesters</option></select>
      </div>
      <div id="material-results" class="grid"></div>
    </section>
    <section id="page-dashboard" class="page hidden">
      <div class="section-heading"><div><div class="eyebrow">Your account</div><h2>My library</h2><p id="account-greeting"></p></div></div>
      <div class="card" style="margin-bottom:16px"><h3>My profile</h3><div id="profile-details"></div></div>
      <div class="card"><h3>Borrow history</h3><p>Current overdue fines are calculated at $1 per day.</p><div id="history-table" class="table-wrap"></div></div>
    </section>
    <section id="page-admin" class="page hidden">
      <div class="section-heading"><div><div class="eyebrow">Library management</div><h2>Librarian dashboard</h2><p>Manage books, loans, and study materials.</p></div></div>
      <div class="stack">
        <div class="card"><h3>Add a book</h3>
          <form id="book-form">
            <input type="hidden" name="bookId">
            <div class="form-grid"><label>Book title<input name="title" required></label><label>Author<input name="author" required></label><label>Subject<input name="subject" required></label><label>Copies<input name="copies" type="number" min="1" value="1" required></label></div>
            <div><button class="button" type="submit">Add book</button> <button class="button secondary hidden" id="cancel-book-edit" type="button">Cancel edit</button></div>
          </form>
        </div>
        <div class="card"><h3>Inventory</h3><div id="inventory-table" class="table-wrap"></div></div>
        <div class="card"><h3>Issue a book</h3>
          <form id="issue-form">
            <div class="form-grid"><label>Student account<select name="user" id="student-select" required></select></label><label>Available book<select name="book" id="issue-book-select" required></select></label><label>Expected return date<input name="due" type="date" required></label></div>
            <button class="button" type="submit">Issue book</button>
          </form>
        </div>
        <div class="card"><h3>Issued books &amp;amp; returns</h3><div id="loans-table" class="table-wrap"></div></div>
        <div class="card"><h3>Add study material</h3>
          <form id="material-form">
```

```
            <input type="hidden" name="materialId">
            <div class="form-grid"><label>Material title<input name="title"
required></label><label>Year<input name="year" type="number" min="1" max="8"
placeholder="e.g. 1" required></label><label>Semester<input name="semester"
type="number" min="1" max="12" placeholder="e.g. 1" required></label><label>File<input
name="file" type="file" accept=".pdf,.doc,.docx,.ppt,.pptx,.txt"></label></div>
            <div><button class="button" type="submit">Add material</button> <button
class="button secondary hidden" id="cancel-material-edit" type="button">Cancel
edit</button></div>
          </form>
          <div id="admin-materials" class="table-wrap" style="margin-top:18px"></div>
        </div>
      </div>
    </section>
    <section id="page-contact" class="page hidden">
      <div class="section-heading"><div><div class="eyebrow">Get in
touch</div><h2>Contact the library</h2><p>Send the librarian a
message.</p></div></div>
      <div class="card" style="max-width:620px">
        <form id="contact-form">
          <label>Your name<input name="name" required></label><label>Email<input
type="email" name="email" required></label><label>Message<textarea name="message"
rows="5" required></textarea></label>
          <button class="button" type="submit">Send message</button>
        </form>
      </div>
    </section>
  </main>
  <footer>Cornerstone Library · Open for your next chapter</footer>
  <div id="auth-modal" class="modal hidden" role="dialog" aria-modal="true" aria-
labelledby="auth-title">
    <div class="modal-card">
      <div class="modal-head"><h2 id="auth-title">Log in</h2><button class="close"
id="modal-close" aria-label="Close">×</button></div>
      <div class="toolbar"><button class="button small" id="show-login">Log
in</button><button class="button secondary small" id="show-register">Create
account</button></div>
      <form id="login-form"><label>Email<input name="email" type="email" required
autocomplete="username"></label><label>Password<input name="password" type="password"
required autocomplete="current-password"></label><button class="button"
type="submit">Log in</button></form>
      <form id="register-form" class="hidden"><label>Name<input name="name" required
autocomplete="name"></label><label>Email<input name="email" type="email" required
autocomplete="email"></label><label>Password<input name="password" type="password"
minlength="4" required autocomplete="new-password"></label><button class="button"
type="submit">Create account</button></form>
      <p style="font-size:.82rem;margin-bottom:0">Demo librarian:
librarian@library.test / library123</p>
```

```
    </div>
  </div>
  <script>
    const KEY = "cornerstone-library-v1";
    const initial = {
      users: [
        { name: "Library Admin", email: "librarian@library.test", password:
"library123", role: "admin" }
      ],
      books: [
        { id: 1, title: "The Great Gatsby", author: "F. Scott Fitzgerald", subject:
"Literature", copies: 3 },
        { id: 2, title: "A Brief History of Time", author: "Stephen Hawking", subject:
"Science", copies: 2 },
        { id: 3, title: "Introduction to Algorithms", author: "Thomas H. Cormen",
subject: "Computer Science", copies: 4 },
        { id: 4, title: "Pride and Prejudice", author: "Jane Austen", subject:
"Literature", copies: 2 },
        { id: 5, title: "The Selfish Gene", author: "Richard Dawkins", subject:
"Biology", copies: 1 },
        { id: 6, title: "The Art of Statistics", author: "David Spiegelhalter",
subject: "Mathematics", copies: 2 }
      ],
      loans: [],
      materials: [
        { id: 1, title: "Introduction to Biology – Notes", year: "1", semester: "1",
filename: "biology-notes.txt", data: "Welcome to the library study materials. Replace
this sample with your course notes." },
        { id: 2, title: "Calculus – Revision Guide", year: "1", semester: "2",
filename: "calculus-guide.txt", data: "Sample revision guide. Librarians can add real
course files from the dashboard." },
        { id: 3, title: "Database Systems – Lecture Notes", year: "2", semester: "1",
filename: "database-notes.txt", data: "Sample lecture notes. Upload a PDF or document
in the librarian dashboard." }
      ]
    };
    let db;
    try {
      db = JSON.parse(localStorage.getItem(KEY)) || initial;
      if (!Array.isArray(db.users) || !Array.isArray(db.books) ||
!Array.isArray(db.loans) || !Array.isArray(db.materials)) throw new Error("Invalid
saved library data");
    } catch (error) {
      console.error("Could not load saved library data:", error);
      db = JSON.parse(JSON.stringify(initial));
    }
    let currentUser = null;
    const $ = (selector) => document.querySelector(selector);
```

```
    const escapeHTML = (value) => String(value).replace(/[&<>"']/g, c => ({ "&": "&amp;", "<": "&lt;", ">": "&gt;", '"': "&quot;", "'": "&#39;" })[c]);
    function save() { localStorage.setItem(KEY, JSON.stringify(db)); }
    function notice(text, error = false) {
      const el = $("#message"); el.textContent = text; el.classList.remove("hidden", "error");
      if (error) el.classList.add("error");
      window.scrollTo({ top: 0, behavior: "smooth" });
      setTimeout(() => el.classList.add("hidden"), 4500);
    }
    function available(book) {
      return book.copies - db.loans.filter(loan => loan.bookId === book.id && !loan.returned).length;
    }
    function fine(loan) {
      if (loan.returned) return 0;
      const days = Math.floor((Date.now() - new Date(loan.due + "T23:59:59").getTime()) / 86400000);
      return Math.max(0, days) * 1;
    }
    function go(page) {
      document.querySelectorAll(".page").forEach(section => section.classList.toggle("hidden", section.id !== `page-${page}`));
      document.querySelectorAll("nav [data-page]").forEach(button => button.classList.toggle("active", button.dataset.page === page));
      if (page === "catalog") renderBooks();
      if (page === "materials") renderMaterials();
      if (page === "dashboard") renderDashboard();
      if (page === "admin") renderAdmin();
      window.scrollTo(0, 0);
    }
    function bookCard(book) {
      const count = available(book);
      return `<article class="card book-card"><h3>${escapeHTML(book.title)}</h3><div class="meta">by ${escapeHTML(book.author)}</div><div class="meta">${escapeHTML(book.subject)}</div><div class="availability"><span class="badge ${count < 1 ? "unavailable" : ""}">${count} available</span><button class="button secondary small details-button" data-id="${book.id}">Details</button></div></article>`;
    }
    function renderBooks() {
      const query = ($("#book-search").value || "").trim().toLowerCase();
      const books = db.books.filter(book => `${book.title} ${book.author} ${book.subject}`.toLowerCase().includes(query));
      $("#book-results").innerHTML = books.length ? books.map(bookCard).join("") : '<div class="empty">No books found. Try another search.</div>';
      document.querySelectorAll(".details-button").forEach(button => button.addEventListener("click", () => {
```

```
        const book = db.books.find(item => item.id === Number(button.dataset.id));
        if (book) notice(`${book.title} by ${book.author} · ${book.subject} · ${available(book)} copies available.`);
      }));
    }
    function renderMaterials() {
      const years = [...new Set(db.materials.map(item => item.year))].sort((a,b) => Number(a)-Number(b));
      const semesters = [...new Set(db.materials.map(item => item.semester))].sort((a,b) => Number(a)-Number(b));
      const year = $("#material-year").value, semester = $("#material-semester").value;
      $("#material-year").innerHTML = '<option value="">All years</option>' + years.map(value => `<option value="${escapeHTML(value)}">${escapeHTML(value)}</option>`).join("");
      $("#material-semester").innerHTML = '<option value="">All semesters</option>' + semesters.map(value => `<option value="${escapeHTML(value)}">${escapeHTML(value)}</option>`).join("");
      $("#material-year").value = year; $("#material-semester").value = semester;
      const filtered = db.materials.filter(item => (!year || item.year === year) && (!semester || item.semester === semester));
      $("#material-results").innerHTML = filtered.length ? filtered.map(item => `<article class="card"><div class="eyebrow">Year ${escapeHTML(item.year)} · Semester ${escapeHTML(item.semester)}</div><h3 style="margin-top:10px">${escapeHTML(item.title)}</h3><a class="button secondary small" style="display:inline-block;text-decoration:none;margin-top:10px" href="${item.data || ""}" download="${escapeHTML(item.filename)}">Download material</a></article>`).join("") : '<div class="empty">No materials match these filters.</div>';
    }
    function renderDashboard() {
      if (!currentUser || currentUser.role !== "user") return go("home");
      $("#account-greeting").textContent = `Welcome back, ${currentUser.name}.`;
      $("#profile-details").innerHTML = `<p><strong>Name:</strong> ${escapeHTML(currentUser.name)}<br><strong>Email:</strong> ${escapeHTML(currentUser.email)}</p>`;
      const loans = db.loans.filter(loan => loan.email === currentUser.email).sort((a,b) => b.issue.localeCompare(a.issue));
      $("#history-table").innerHTML = loans.length ? `<table><thead><tr><th>Book</th><th>Issue date</th><th>Due date</th><th>Status</th><th>Fine</th></tr></thead><tbody>${loans.map(loan => `<tr><td>${escapeHTML(db.books.find(book => book.id === loan.bookId)?.title || "Book removed")}</td><td>${escapeHTML(loan.issue)}</td><td>${escapeHTML(loan.due)}</td><td>${loan.returned ? "Returned" : (fine(loan) ? "Overdue" : "Issued")}</td><td class="${fine(loan) ? "fine" : ""}">$${fine(loan)}</td></tr>`).join("")}</tbody></table>` : '<div class="empty">No borrowed books yet.</div>';
    }
```

```
    function renderAdmin() {
      if (!currentUser || currentUser.role !== "admin") return go("home");
      $("#inventory-table").innerHTML =
`<table><thead><tr><th>Book</th><th>Author</th><th>Copies</th><th>Available</th><th>Ac
tion</th></tr></thead><tbody>${db.books.map(book =>
`<tr><td>${escapeHTML(book.title)}</td><td>${escapeHTML(book.author)}</td><td>${book.c
opies}</td><td>${available(book)}</td><td><button class="button secondary small edit-
book" data-id="${book.id}">Edit</button> <button class="button danger small delete-
book" data-id="${book.id}">Delete</button></td></tr>`).join("") || '<tr><td
colspan="5">No books in inventory.</td></tr>'}</tbody></table>`;
      document.querySelectorAll(".edit-book").forEach(button =>
button.addEventListener("click", () => {
        const book = db.books.find(item => item.id === Number(button.dataset.id));
        if (!book) return;
        const form = $("#book-form");
        form.elements.bookId.value = book.id;
        form.elements.title.value = book.title;
        form.elements.author.value = book.author;
        form.elements.subject.value = book.subject;
        form.elements.copies.min = book.copies - available(book);
        form.elements.copies.value = book.copies;
        form.querySelector('[type="submit"]').textContent = "Update book";
        $("#cancel-book-edit").classList.remove("hidden");
        form.scrollIntoView({ behavior: "smooth", block: "center" });
      }));
      document.querySelectorAll(".delete-book").forEach(button =>
button.addEventListener("click", () => {
        const id = Number(button.dataset.id);
        if (db.loans.some(loan => loan.bookId === id && !loan.returned)) return
notice("Return all copies before deleting this book.", true);
        db.books = db.books.filter(book => book.id !== id); save(); renderAdmin();
renderBooks();
      }));
      const students = db.users.filter(user => user.role === "user");
      $("#student-select").innerHTML = students.map(user => `<option
value="${escapeHTML(user.email)}">${escapeHTML(user.name)}
(${escapeHTML(user.email)})</option>`).join("");
      $("#issue-book-select").innerHTML = db.books.filter(book => available(book) >
0).map(book => `<option value="${book.id}">${escapeHTML(book.title)}
(${available(book)} available)</option>`).join("");
      $("#issue-form button[type=submit]").disabled = !students.length ||
!db.books.some(book => available(book) > 0);
      const loans = [...db.loans].sort((a,b) => b.issue.localeCompare(a.issue));
      $("#loans-table").innerHTML = loans.length ?
`<table><thead><tr><th>Student</th><th>Book</th><th>Issued</th><th>Due</th><th>Fine</t
h><th>Status</th></tr></thead><tbody>${loans.map(loan =>
`<tr><td>${escapeHTML(loan.email)}</td><td>${escapeHTML(db.books.find(book => book.id
=== loan.bookId)?.title || "Book
```

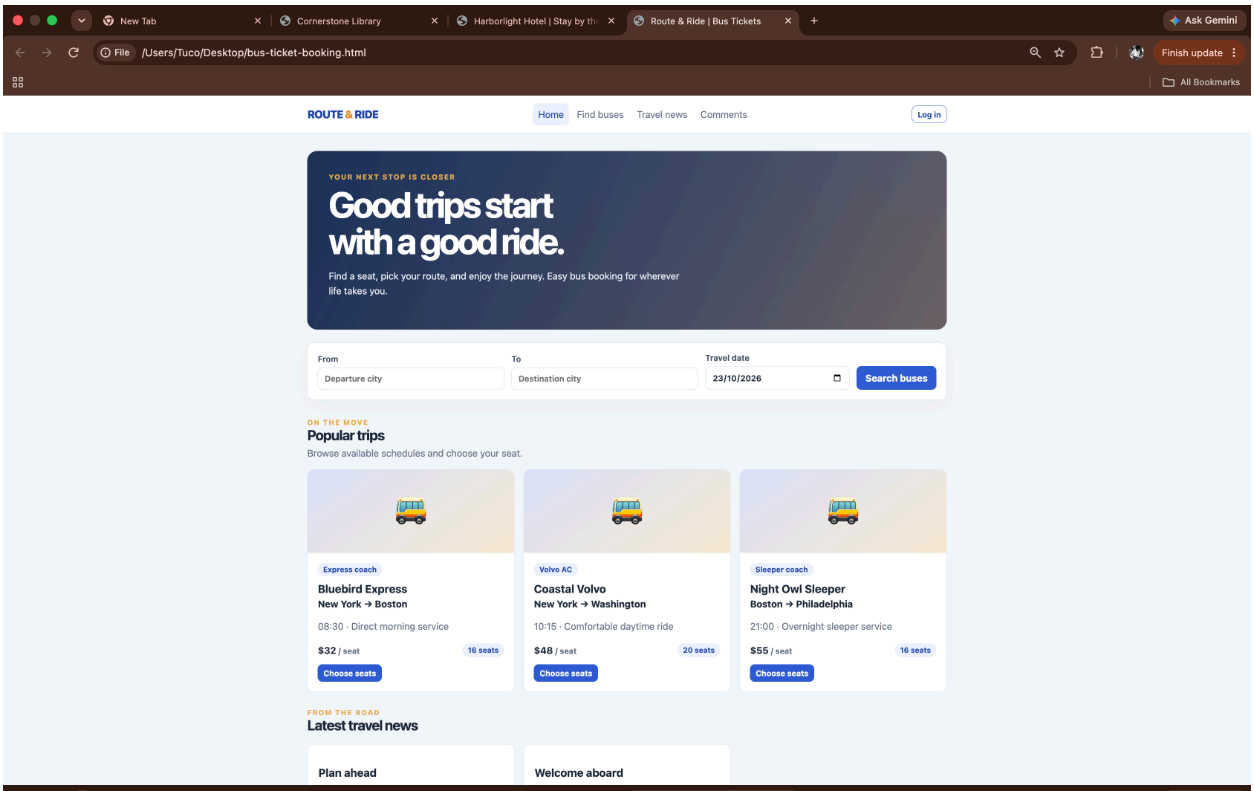
```
removed")}</td><td>${escapeHTML(loan.issue)}</td><td>${escapeHTML(loan.due)}</td><td
class="${fine(loan) ? "fine" : ""}">$${fine(loan)}</td><td>${loan.returned ?
"Returned" : `<button class="button secondary small return-book"
data-id="${loan.id}">Mark returned</button>`}</td></tr>`).join("")}</tbody></table>` :
'<div class="empty">No books have been issued.</div>';
      document.querySelectorAll(".return-book").forEach(button =>
button.addEventListener("click", () => {
        const loan = db.loans.find(item => item.id === Number(button.dataset.id));
        if (loan) { loan.returned = true; loan.returnedOn = new
Date().toISOString().slice(0,10); save(); renderAdmin(); }
      }));
      $("#admin-materials").innerHTML = db.materials.length ?
`<table><thead><tr><th>Material</th><th>Year</th><th>Semester</th><th>Actions</th></tr
></thead><tbody>${db.materials.map(item =>
`<tr><td>${escapeHTML(item.title)}</td><td>${escapeHTML(item.year)}</td><td>${escapeHT
ML(item.semester)}</td><td><button class="button secondary small edit-material" data-
id="${item.id}">Edit</button> <button class="button danger small delete-material"
data-id="${item.id}">Delete</button></td></tr>`).join("")}</tbody></table>` : '<div
class="empty">No study materials added.</div>';
      document.querySelectorAll(".edit-material").forEach(button =>
button.addEventListener("click", () => {
        const material = db.materials.find(item => item.id ===
Number(button.dataset.id));
        if (!material) return;
        const form = $("#material-form");
        form.elements.materialId.value = material.id;
        form.elements.title.value = material.title;
        form.elements.year.value = material.year;
        form.elements.semester.value = material.semester;
        form.querySelector('[type="submit"]').textContent = "Update material";
        $("#cancel-material-edit").classList.remove("hidden");
        form.scrollIntoView({ behavior: "smooth", block: "center" });
      }));
      document.querySelectorAll(".delete-material").forEach(button =>
button.addEventListener("click", () => {
        db.materials = db.materials.filter(item => item.id !==
Number(button.dataset.id));
        save(); renderAdmin(); renderMaterials();
      }));
    }
    function updateAuth() {
      const loggedIn = !!currentUser;
      $("#login-open").classList.toggle("hidden", loggedIn);
      $("#logout").classList.toggle("hidden", !loggedIn);
      $("#user-label").textContent = loggedIn ? currentUser.name : "";
      $("#dashboard-nav").classList.toggle("hidden", !loggedIn || currentUser.role !==
"user");
```

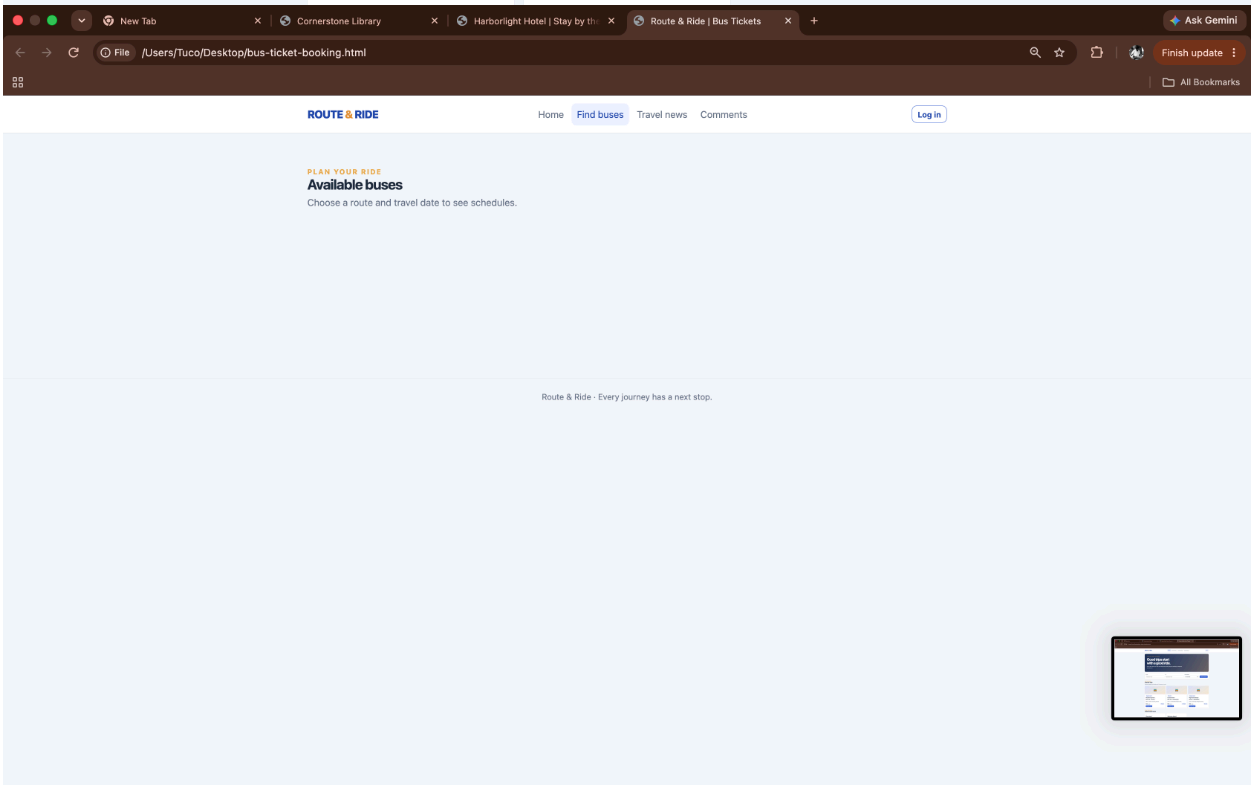
```
    $("#admin-nav").classList.toggle("hidden", !loggedIn || currentUser.role !== "admin");
  }
  document.querySelectorAll("nav [data-page]").forEach(button => button.addEventListener("click", () => go(button.dataset.page)));
  document.querySelectorAll("[data-go]").forEach(button => button.addEventListener("click", () => go(button.dataset.go)));
  $("#book-search").addEventListener("input", renderBooks);
  $("#material-year").addEventListener("change", renderMaterials);
  $("#material-semester").addEventListener("change", renderMaterials);
  $("#login-open").addEventListener("click", () => $("#auth-modal").classList.remove("hidden"));
  $("#modal-close").addEventListener("click", () => $("#auth-modal").classList.add("hidden"));
  $("#auth-modal").addEventListener("click", event => { if (event.target.id === "auth-modal") $("#auth-modal").classList.add("hidden"); });
  $("#show-login").addEventListener("click", () => { $("#login-form").classList.remove("hidden"); $("#register-form").classList.add("hidden"); $("#auth-title").textContent = "Log in"; });
  $("#show-register").addEventListener("click", () => { $("#register-form").classList.remove("hidden"); $("#login-form").classList.add("hidden"); $("#auth-title").textContent = "Create account"; });
  $("#login-form").addEventListener("submit", event => {
    event.preventDefault();
    const data = new FormData(event.currentTarget), email = String(data.get("email")).trim().toLowerCase();
    const user = db.users.find(item => item.email === email && item.password === data.get("password"));
    if (!user) return notice("Email or password is incorrect.", true);
    currentUser = user; updateAuth(); $("#auth-modal").classList.add("hidden"); event.currentTarget.reset();
    go(user.role === "admin" ? "admin" : "dashboard"); notice(`Signed in as ${user.name}.`);
  });
  $("#register-form").addEventListener("submit", event => {
    event.preventDefault();
    const data = new FormData(event.currentTarget), email = String(data.get("email")).trim().toLowerCase();
    if (db.users.some(user => user.email === email)) return notice("An account with this email already exists.", true);
    const user = { name: String(data.get("name")).trim(), email, password: String(data.get("password")), role: "user" };
    db.users.push(user); save(); currentUser = user; updateAuth(); $("#auth-modal").classList.add("hidden"); event.currentTarget.reset();
    go("dashboard"); notice("Your account is ready.");
  });
```

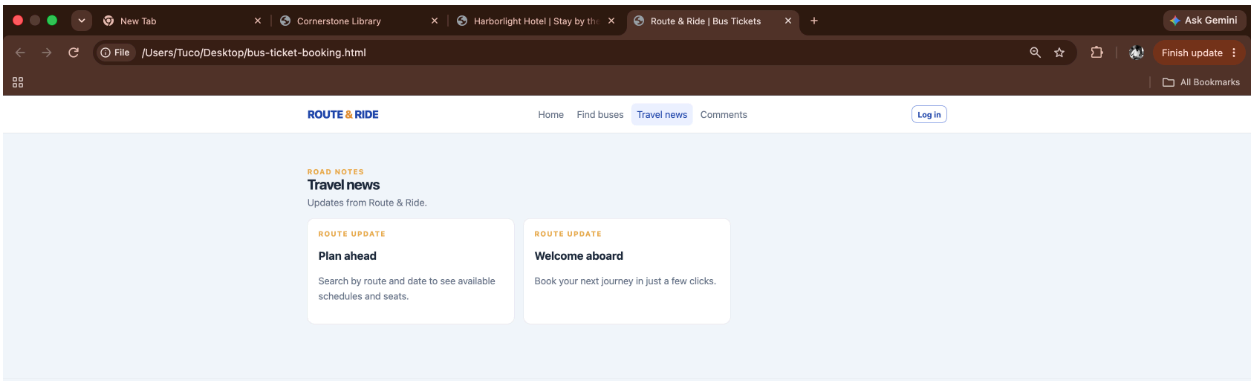
```
    $("#logout").addEventListener("click", () => { currentUser = null; updateAuth();
go("home"); notice("You have logged out."); });
    $("#book-form").addEventListener("submit", event => {
      event.preventDefault(); const data = new FormData(event.currentTarget);
      const id = Number(data.get("bookId"));
      const book = db.books.find(item => item.id === id);
      if (book) {
        const borrowed = book.copies - available(book);
        if (Number(data.get("copies")) < borrowed) return notice(`Copies cannot be
lower than the ${borrowed} currently issued.`, true);
        Object.assign(book, { title: String(data.get("title")).trim(), author:
String(data.get("author")).trim(), subject: String(data.get("subject")).trim(),
copies: Number(data.get("copies")) });
      } else {
        db.books.push({ id: Date.now(), title: String(data.get("title")).trim(),
author: String(data.get("author")).trim(), subject:
String(data.get("subject")).trim(), copies: Number(data.get("copies")) });
      }
      save(); event.currentTarget.reset(); event.currentTarget.elements.copies.min =
1; event.currentTarget.querySelector('[type="submit"]').textContent = "Add book";
$("#cancel-book-edit").classList.add("hidden"); renderAdmin(); renderBooks();
notice(book ? "Book updated." : "Book added to inventory.");
    });
    $("#cancel-book-edit").addEventListener("click", () => {
      const form = $("#book-form"); form.reset(); form.elements.copies.min = 1;
form.querySelector('[type="submit"]').textContent = "Add book";
$("#cancel-book-edit").classList.add("hidden");
    });
    $("#issue-form").addEventListener("submit", event => {
      event.preventDefault(); const data = new FormData(event.currentTarget), bookId =
Number(data.get("book"));
      const book = db.books.find(item => item.id === bookId);
      if (!book || available(book) < 1) return notice("No copies of this book are
currently available.", true);
      db.loans.push({ id: Date.now(), email: String(data.get("user")), bookId, issue:
new Date().toISOString().slice(0,10), due: String(data.get("due")), returned: false
});
      save(); renderAdmin(); notice("Book issued successfully.");
    });
    $("#material-form").addEventListener("submit", event => {
      event.preventDefault();
      const data = new FormData(event.currentTarget), file = data.get("file");
      const id = Number(data.get("materialId"));
      const material = db.materials.find(item => item.id === id);
      const finish = (fileData, filename) => {
        const values = { title: String(data.get("title")).trim(), year:
String(data.get("year")), semester: String(data.get("semester")) };
```

```
        if (material) Object.assign(material, values, fileData === null ? {} : { data:
fileData, filename });
        else db.materials.push({ id: Date.now(), ...values, filename, data: fileData
});
        save(); event.currentTarget.reset();
event.currentTarget.querySelector('[type="submit"]').textContent = "Add material";
$("#cancel-material-edit").classList.add("hidden"); renderAdmin(); renderMaterials();
notice(material ? "Study material updated." : "Study material added.");
      };
      if (file instanceof File && file.size) {
        const reader = new FileReader();
        reader.onerror = () => notice("Could not read the selected material file.",
true);
        reader.onload = () => finish(reader.result, file.name);
        reader.readAsDataURL(file);
      } else if (material) finish(null, material.filename);
      else notice("Choose a file to add the study material.", true);
    });
    $("#cancel-material-edit").addEventListener("click", () => {
      const form = $("#material-form"); form.reset();
form.querySelector('[type="submit"]').textContent = "Add material"; $("#cancel-
material-edit").classList.add("hidden");
    });
    $("#contact-form").addEventListener("submit", event => {
      event.preventDefault(); event.currentTarget.reset();
      notice("Thanks for your message. In this demo, connect the form to an email
service to deliver it.");
    });
    $("#featured-books").innerHTML = db.books.slice(0,3).map(bookCard).join("");
    renderBooks(); renderMaterials(); updateAuth();
  </script>
</body>
</html>
```

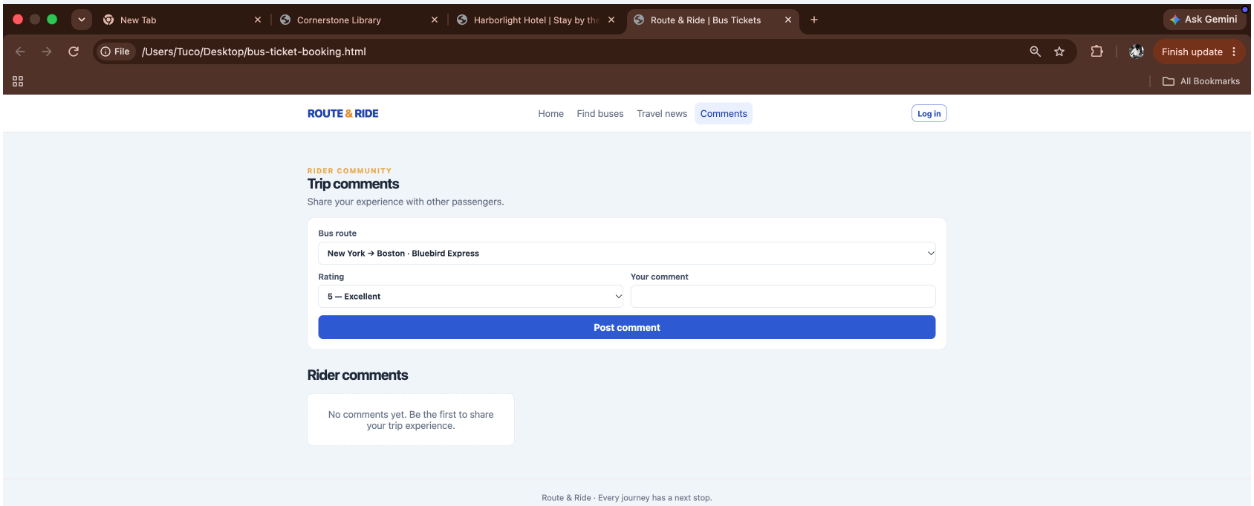
OUTPUT:


New Tab
Cornerstone Library
Harborlight Hotel | Stay by the
Route & Ride | Bus Tickets
Ask Gemini
File /Users/Tuco/Desktop/bus-ticket-booking.html
Finish update
All Bookmarks
ROUTE & RIDE
Home
Find buses
Travel news
Comments
Log in
YOUR NEXT STOP IS CLOSER
Good trips start with a good ride.
Find a seat, pick your route, and enjoy the journey. Easy bus booking for wherever life takes you.
From
Departure city
To
Destination city
Travel date
23/10/2026
Search buses
ON THE MOVE
Popular trips
Browse available schedules and choose your seat.
Express coach
Bluebird Express
New York → Boston
08:30 · Direct morning service
$32 / seat
16 seats
Choose seats
Volvo AC
Coastal Volvo
New York → Washington
10:15 · Comfortable daytime ride
$48 / seat
20 seats
Choose seats
Sleeper coach
Night Owl Sleeper
Boston → Philadelphia
21:00 · Overnight sleeper service
$55 / seat
16 seats
Choose seats
FROM THE ROAD
Latest travel news
Plan ahead
Welcome aboard

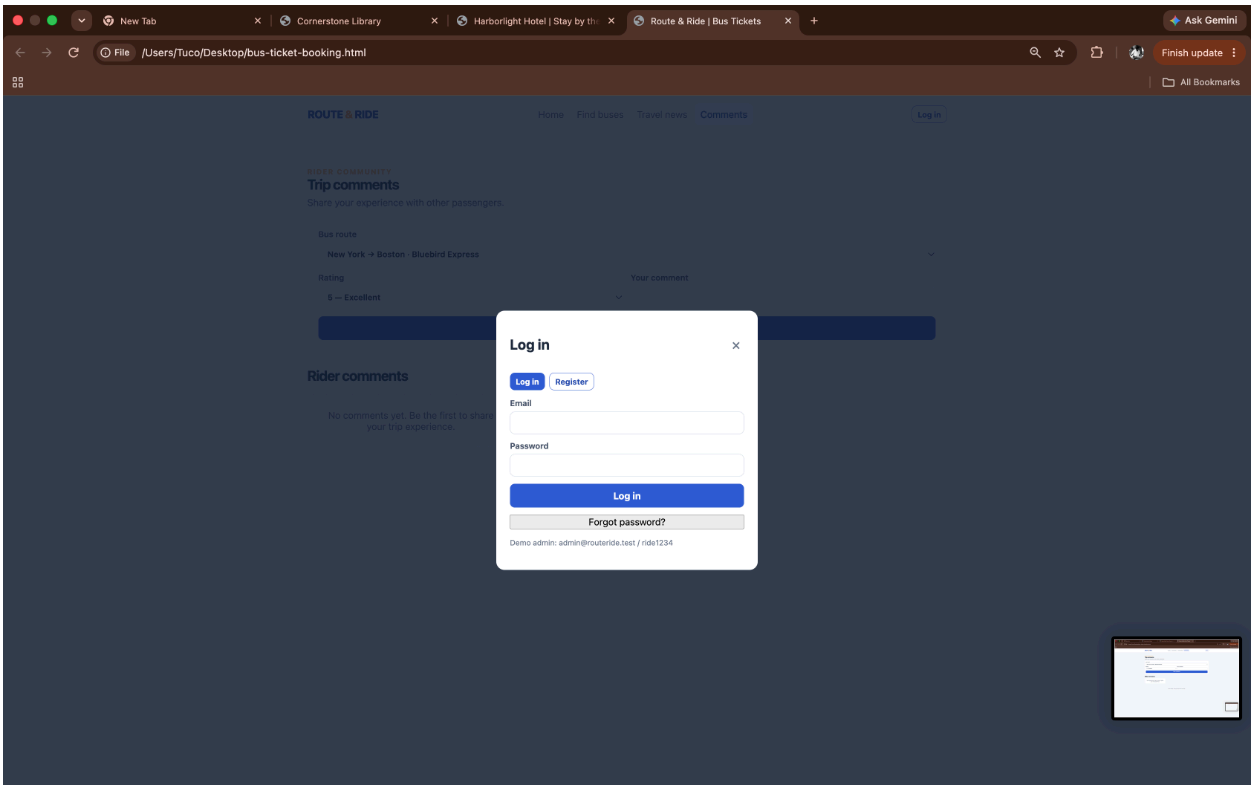

New Tab
Cornerstone Library
Harborlight Hotel | Stay by the
Route & Ride | Bus Tickets
Ask Gemini
File /Users/Tuco/Desktop/bus-ticket-booking.html
Finish update
All Bookmarks
ROUTE & RIDE
Home
Find buses
Travel news
Comments
Log in
PLAN YOUR RIDE
Available buses
Choose a route and travel date to see schedules.
Route & Ride · Every journey has a next stop.


New Tab
Cornerstone Library
Harborlight Hotel | Stay by the
Route & Ride | Bus Tickets
Ask Gemini
File /Users/Tuco/Desktop/bus-ticket-booking.html
Finish update
All Bookmarks
ROUTE & RIDE
Home
Find buses
Travel news
Comments
Log in
ROAD NOTES
Travel news
Updates from Route & Ride.
ROUTE UPDATE
Plan ahead
Search by route and date to see available schedules and seats.
ROUTE UPDATE
Welcome aboard
Book your next journey in just a few clicks.
Route & Ride · Every journey has a next stop.




New Tab
Cornerstone Library
Harborlight Hotel | Stay by the
Route & Ride | Bus Tickets
Ask Gemini
File /Users/Tuco/Desktop/bus-ticket-booking.html
Finish update
All Bookmarks
ROUTE & RIDE
Home
Find buses
Travel news
Comments
Log in
RIDER COMMUNITY
Trip comments
Share your experience with other passengers.
Bus route
New York → Boston · Bluebird Express
Rating
5 — Excellent
Your comment
Post comment
Rider comments
No comments yet. Be the first to share your trip experience.
Route & Ride · Every journey has a next stop.





RESULT : The Online Bus Ticket Booking System was developed to simplify bus reservation and management activities. Users can register, log in, search for buses, view available seats, book and cancel tickets, and post comments about buses. The system allows administrators to add and update bus information, manage bus categories such as Volvo and sleeper coach, maintain registered user details, manage news and comments, and generate reports. The login module supports user authentication, password changes, and password recovery. Thus, the system reduces manual work, improves the management of bus information and bookings, and provides a convenient platform for users and administrators.